

A Reduced-Form Stochastic Intensity Model of Timing Prediction Markets

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Standard Prediction Markets

- "Will event A occur at time T ?"
- Iowa Electronic Market (1988)
- \$49B traded on Kalshi and Polymarket in 2025.
- Derivatives of latent sentiment processes (Chen, Ingersoll and Kaplan, 2008)

Will Trump win the 2024 election? \$3.7B traded.



Timing Markets Are Different

- "Will event A occur *before* time T ?"
- Price $\pi_t \rightarrow 0$ as $t \rightarrow T$.
- Can't use same model as outcome markets!
- Previous studies: effect of ousting of Saddam Hussein (Leigh et al., 2003) and Gaddafi (Wörle, 2013) on stock markets

Khamenei Out as Supreme Leader of Iran in 2025?



Reduced-Form Model

- Consider timing market with end date T .
- Event arrives with inhomogeneous Poisson process N_t , having intensity λ_t . Let R be the first-arrival time.
- Yes- vs. no-price

$$\pi_t^{no} = \mathbb{P}[R > T \mid \mathcal{F}_t] = \mathbb{E} \left[e^{-\int_t^T \lambda_s ds} \middle| \mathcal{F}_t \right].$$

- Equivalent to defaultable bonds, and zero-coupon bond with short rate λ_t .

Constant Intensity Baseline

- Simplest model: constant future intensity $\lambda_s \equiv \lambda$ for $s \geq t$

$$\pi_t^{no} = e^{-(T-t)\lambda}$$

- **Instantaneous intensity** invert pricing equation
- **Homogeneous intensity** Least-squares estimate of $\lambda_t \equiv \lambda$ for all t
- **Exponentially weighted intensity** WLSE with decay factor γ
- Estimate using Newton-Raphson

- Price jumps correspond to news events
- JCIR intensity:

$$d\lambda_t = \kappa(\mu - \lambda_t) dt + \sigma\sqrt{\lambda_t} dW_t + dJ_t, \quad \mu, \sigma, \kappa > 0$$

- Compound Poisson jump process J_t . Mean number of jumps α and exponential jump sizes with mean γ . Only upwards!
- Ensures non-negativity
- JCIR-process is affine $\rightarrow$ closed-form solution to pricing equation

$$\pi_t^{no} = \mathbb{E} \left[e^{-\int_t^T \lambda_s ds} \mid \mathcal{F}_t \right] = A(t) e^{-B(t) \lambda_t}.$$

- Identifiability issues!

State-Space Approximation

- No closed-form likelihood. Instead maximize likelihood of Kalman filter innovation residuals. Requires transforming model:

$$\begin{aligned}d\lambda_t &= \kappa(\mu - \lambda_t) dt + \sigma\sqrt{\lambda_t} dW_t + dJ_t \\ \pi_t^{\text{no}} &= A(t)e^{-B(t)\lambda_t}\end{aligned}$$

- Discretize intensity with step size Δ and approximate jumps by moment-matched Gaussian $\rightarrow$ approximate state equation:

$$\lambda_{k+1} = \lambda_k(1 - \kappa\Delta) + \Delta\kappa\mu + \Delta\alpha\gamma + \eta_k, \quad \eta_k \sim \mathcal{N}\left(0, \sigma^2\Delta\lambda_k + 2\Delta\alpha\gamma^2\right).$$

- Add noise to shifted log-price

$$y_k := \log A(t_k) - \log \pi_k^{\text{no}} = B(t_k)\lambda_k + \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(0, s^2).$$

- Almost linear Gaussian state space, but state-dependent variance of η_k . Plug-in estimator!

- Kalman filter produces innovation residuals.

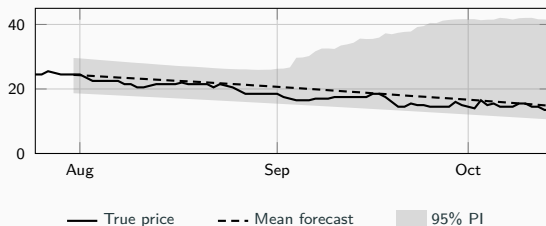
$$e_k = y_k - B(t_k) \mathbb{E}[\lambda_k \mid Y_{k-1}] \rightarrow \text{quasi-likelihood } \ell_{\text{KF}}.$$

- Add probability of market not resolving at each step $\approx 1 - e^{-\Delta\lambda_k} \rightarrow \ell_{\text{Res}}$.
- Estimate parameters by maximizing $\ell_{\text{KF}} + \ell_{\text{Res}}$.
- Kalman filter gives intensity uncertainty $\text{Var}[\lambda_k | y_k, y_{k-1}, \dots]$
- Optimization gives estimate of inverse Hessian. Wald confidence intervals is a rough measure of parameter uncertainty. Unstable!

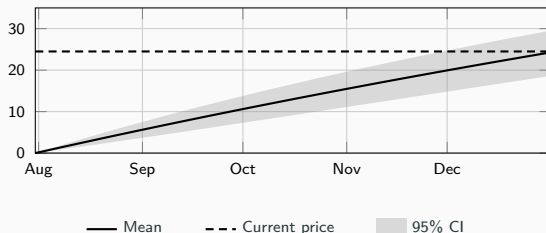
Forecasting Prices and Resolution Probabilities

- Monte Carlo forecast of prices using simulations. Estimates entire predictive distribution
- Account for parameter and intensity uncertainty by drawing from respective normal distributions.
- Probability of event having occurred by any date $s > t$ is simply price of contract with end date s . Calculate explicitly!

Price Forecast

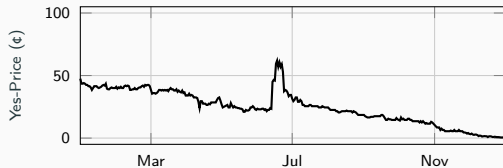


Resolution Probability

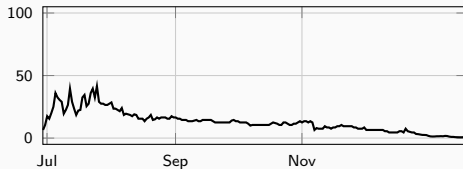


Evaluating on Four Real Timing Markets

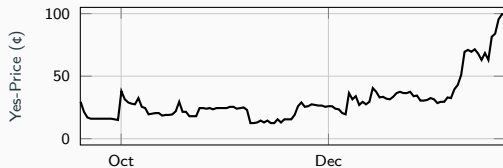
Khamenei Out as Supreme Leader of Iran in 2025?



Will Biden Resign Before the End of his Term?



TikTok Banned in the U.S.
Before May 2025?



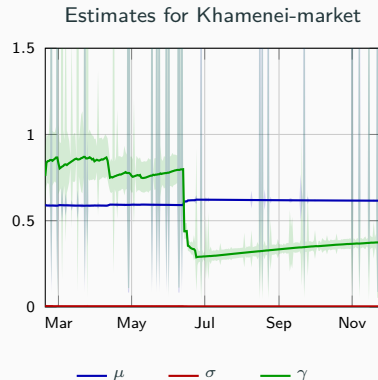
Will Bitcoin hit \$100k in 2024?



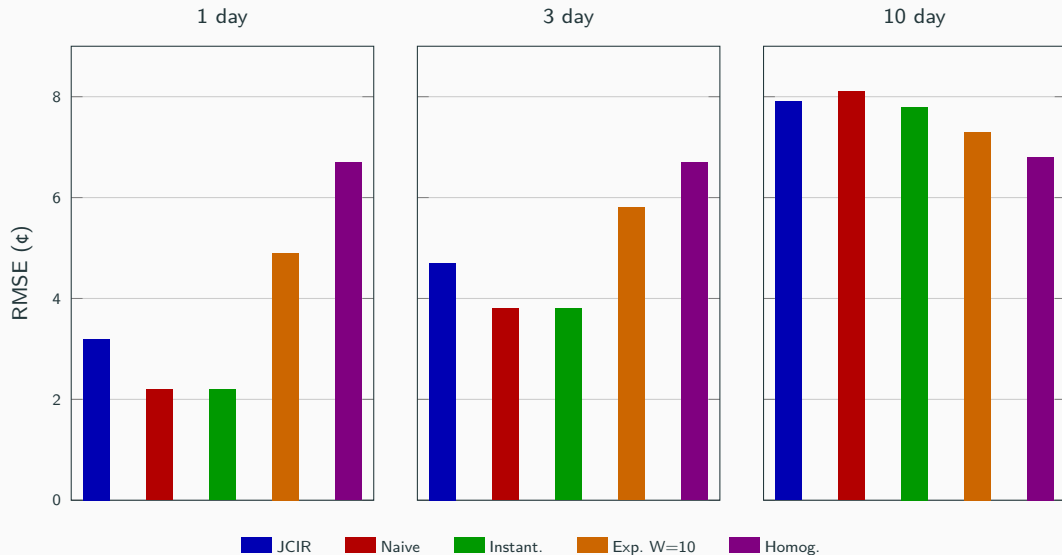
Estimation Setup

- Re-estimate model daily on expanding windows. Min. 50 days. Exclude last 10 % due to numerical issues. Warm start optimizer.
- Weak identification requires holding parameters fixed.

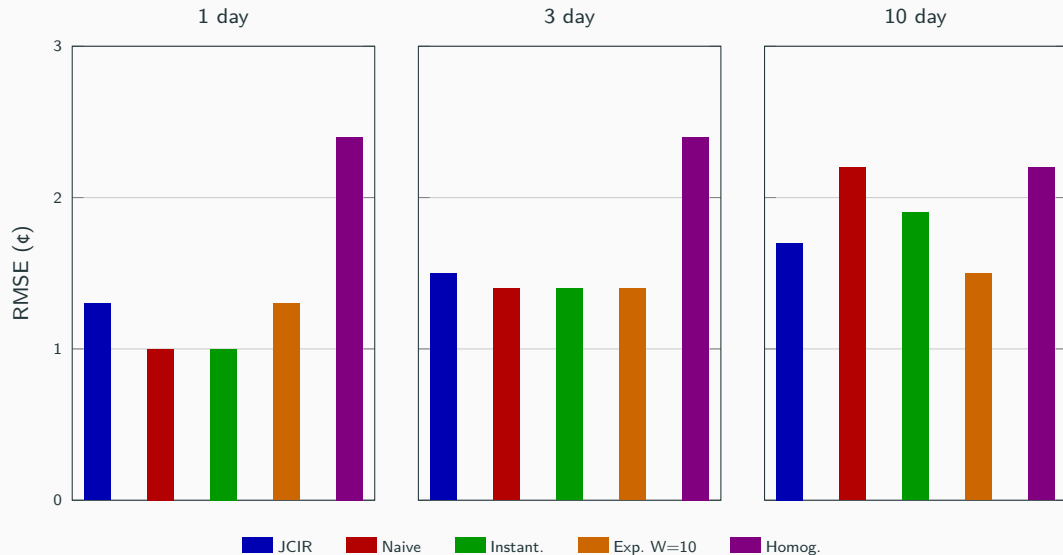
Market	Configuration
Khamenei Out	Fixed $\kappa \equiv 2$, $\alpha \equiv 0.3$
Biden Resignation	Fixed $\kappa \equiv 1$, $\alpha \equiv 0$, $s \equiv 0.01$
TikTok Ban	Fixed $\kappa \equiv 1$, $s \equiv 0.001$
Bitcoin \$100k	Fixed $\kappa \equiv 0.3$, $\alpha \equiv 0$, $s \equiv 0.001$



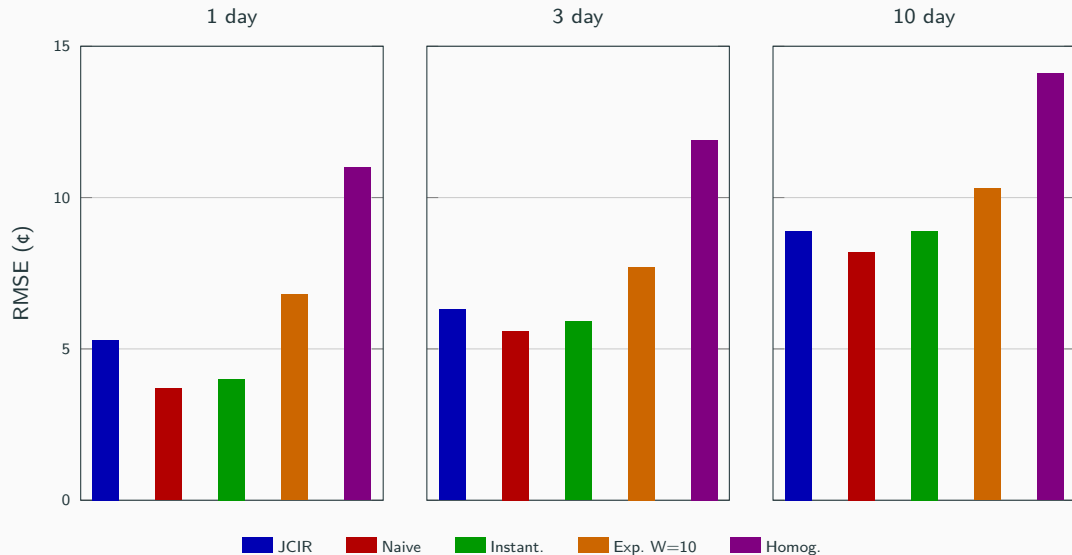
RMSE Results: Khamenei Out



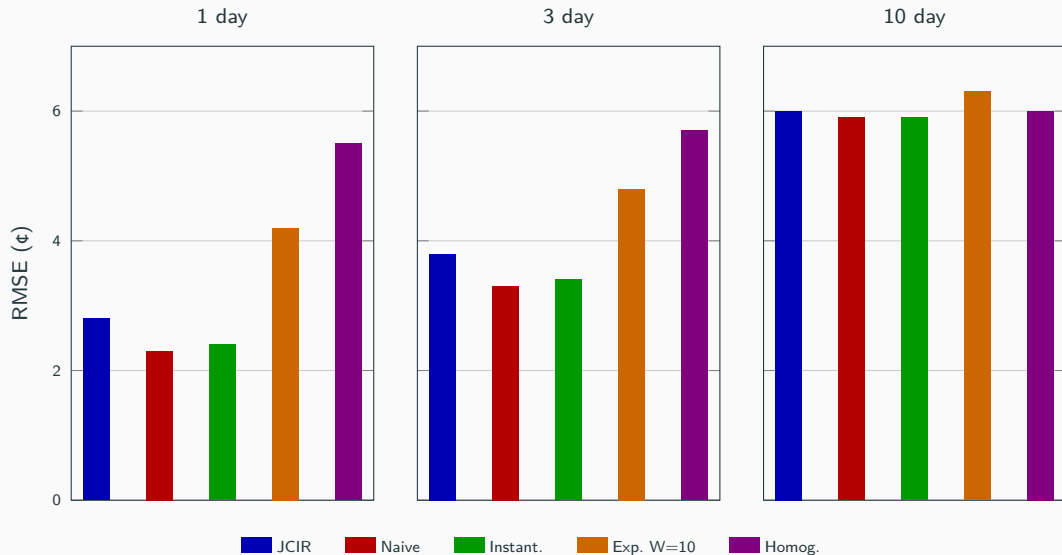
RMSE Results: Biden Resignation



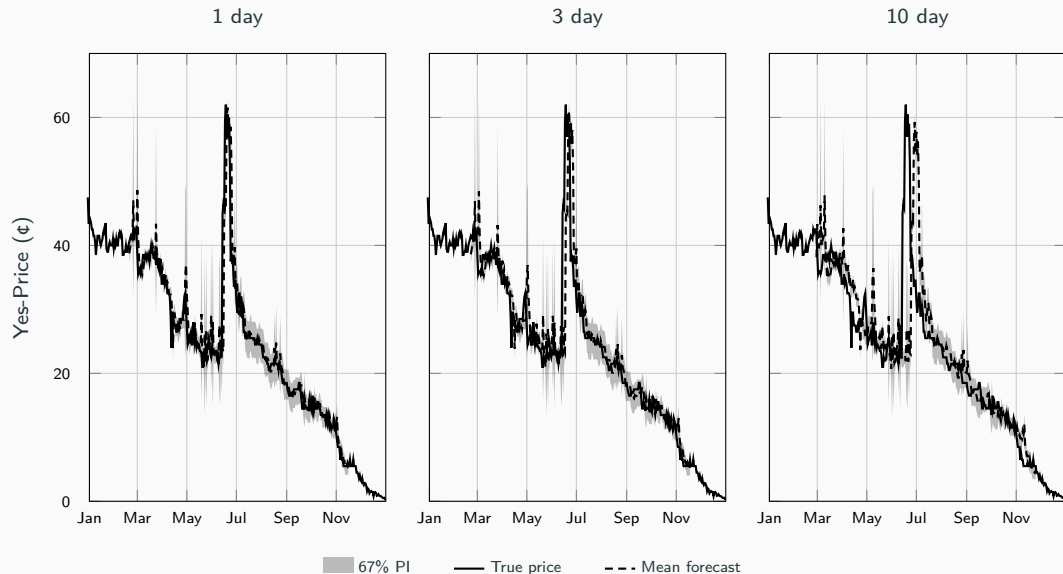
RMSE Results: TikTok Ban



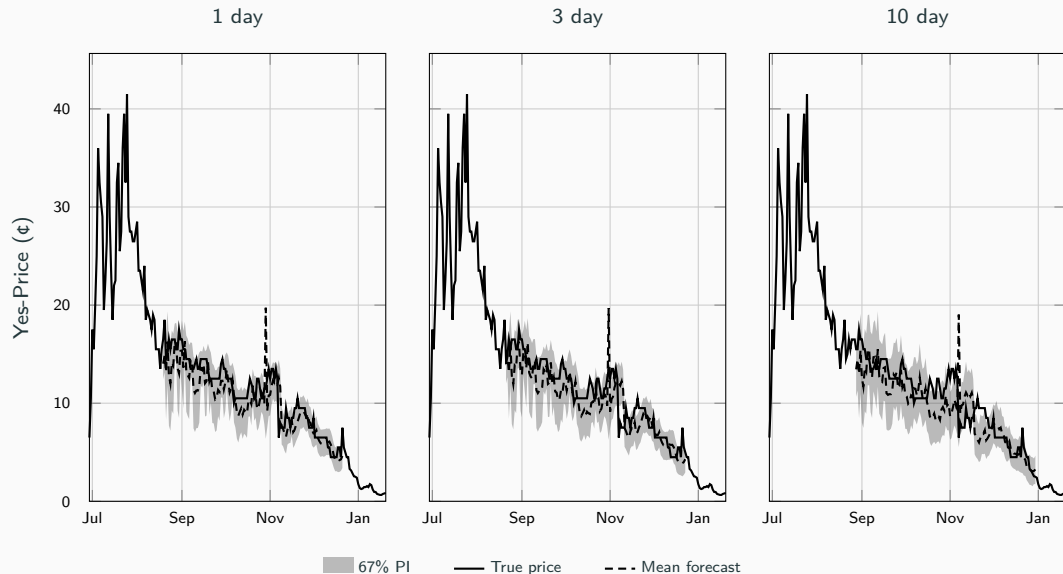
RMSE Results: Bitcoin \$100k



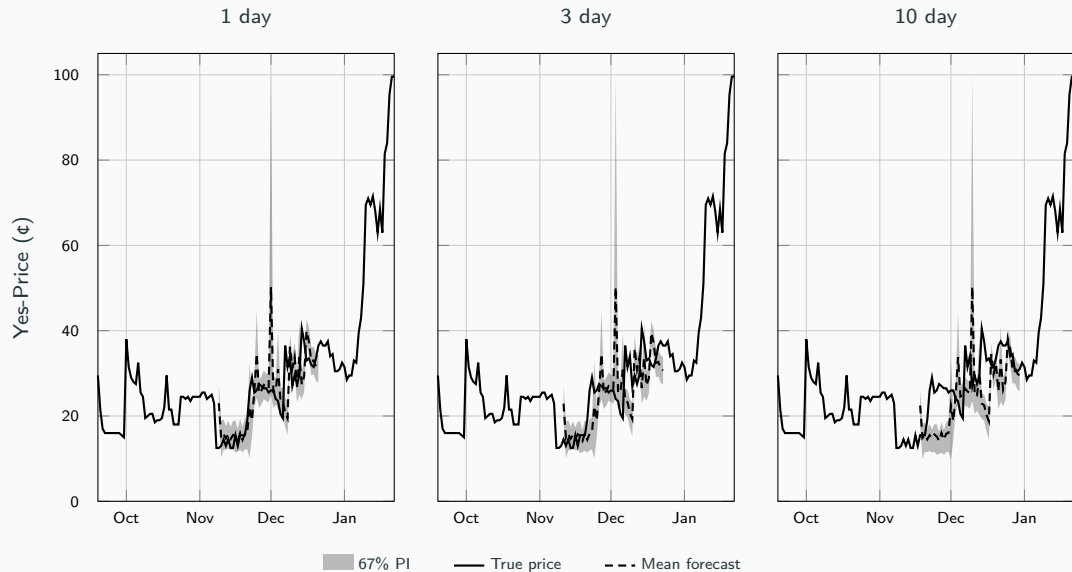
Forecast Paths: Khamenei Out



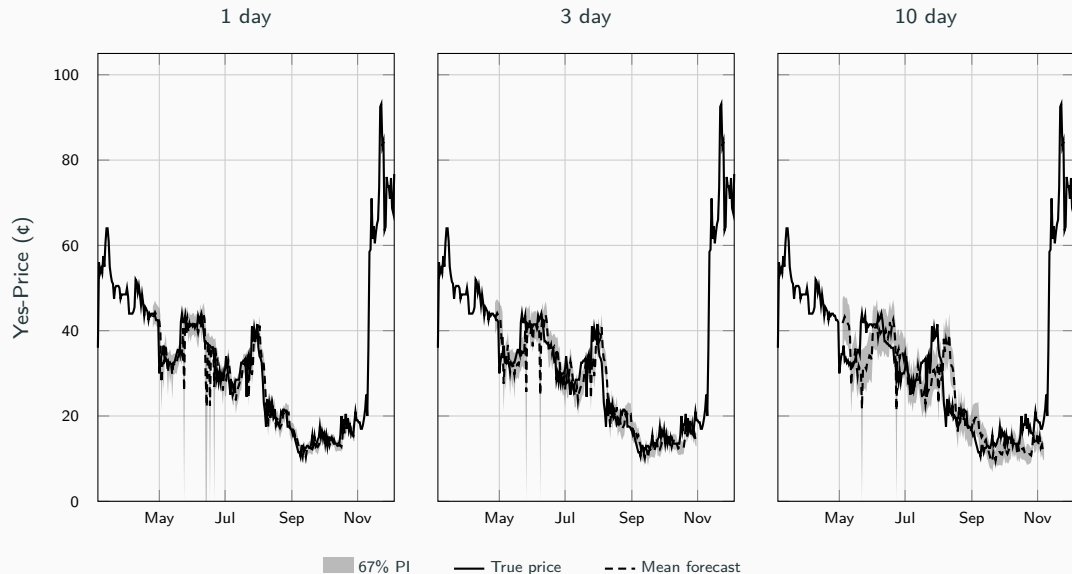
Forecast Paths: Biden Resignation



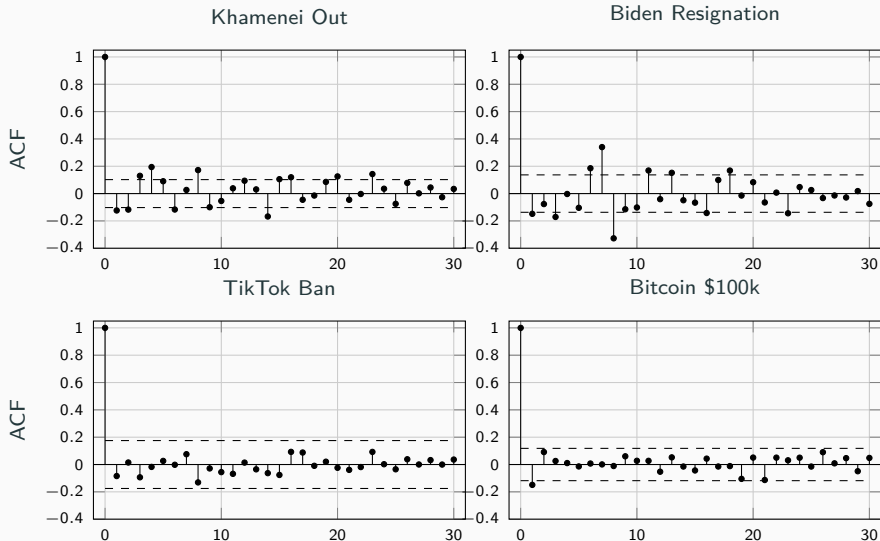
Forecast Paths: TikTok Ban



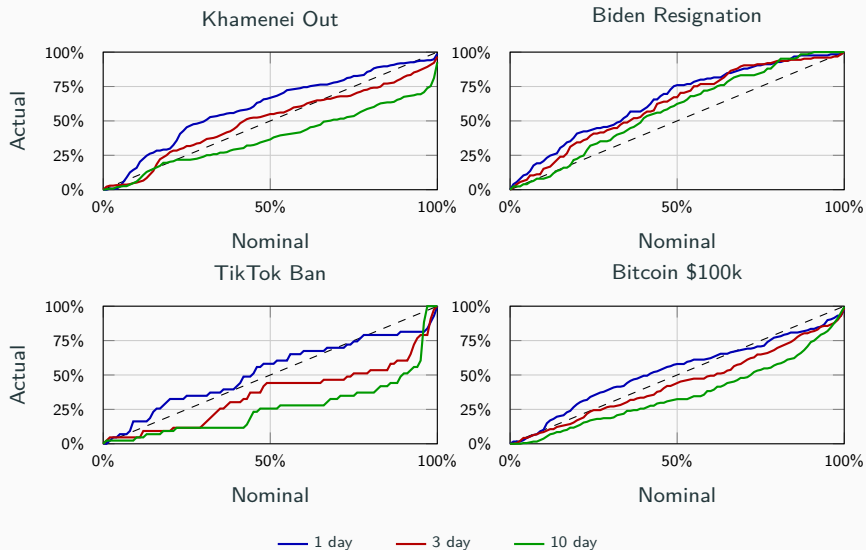
Forecast Paths: Bitcoin \$100k



Little Autocorrelation in Daily Log-Returns



Actual vs. Nominal Coverage Rates



- **Main contribution:** reduced-form pricing framework for timing markets
- JCIR implementation is limited: positive-only jumps, weak identification, unstable uncertainty estimates, and mixed performance.
- Alternate intensity dynamics:
 - Mean-reversion around piecewise constant mean
 - separate large and small jumps, representing different types of news
- Better inference:
 - Parametric Bootstrap
 - Non-linear filters
- Joint estimation across markets with different end dates.

Full Closed-Form Pricing Formula

The no-price under JCIR intensity is

$$\pi_t^{no} = A(t) e^{-B(t) \lambda_t}$$

where

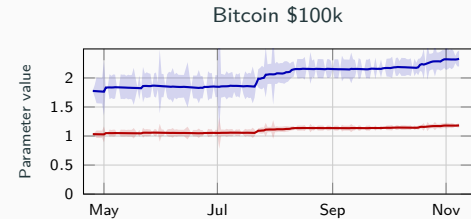
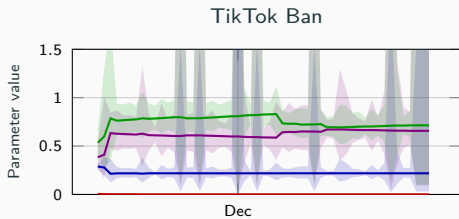
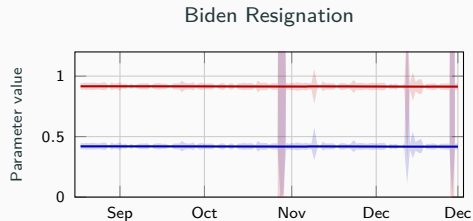
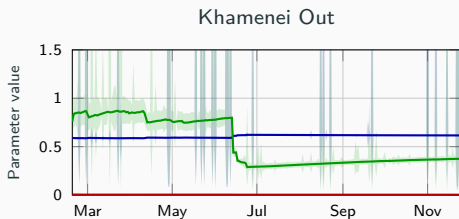
$$h := \sqrt{\kappa^2 + 2\sigma^2}, \quad B(t) := \frac{2 \left(e^{h(T-t)} - 1 \right)}{(\kappa + h) \left(e^{h(T-t)} - 1 \right) + 2h},$$

$$A(t) := \left[\frac{2h e^{(\kappa+h)(T-t)/2}}{(\kappa + h) \left(e^{h(T-t)} - 1 \right) + 2h} \right]^{2\kappa\mu/\sigma^2} \left[\frac{2h e^{(h+\kappa+2\gamma)(T-t)/2}}{2h + (\kappa + h + 2\gamma) \left(e^{h(T-t)} - 1 \right)} \right]^{\frac{2\alpha\gamma}{\sigma^2 - 2\kappa\gamma - 2\gamma^2}}.$$

The denominator $\sigma^2 - 2\kappa\gamma - 2\gamma^2 = [h^2 - (\kappa + 2\gamma)^2] / 2$ is zero when $h^2 = (\kappa + 2\gamma)^2$.
In this limit we have

$$A(t) = \left[\frac{2h e^{(\kappa+h)(T-t)/2}}{(\kappa + h) \left(e^{h(T-t)} - 1 \right) + 2h} \right]^{2\kappa\mu/\sigma^2} \exp \left(-2\alpha\gamma \left[\frac{T-t}{\kappa + h + 2\gamma} + \frac{e^{-h(T-t)} - 1}{h(\kappa + h + 2\gamma)} \right] \right).$$

Parameter Paths for All Four Markets



μ σ α γ

Full RMSE Table

Model	Khamenei			Biden			TikTok			Bitcoin		
	1d	3d	10d	1d	3d	10d	1d	3d	10d	1d	3d	10d
JCIR	3.2	4.7	7.9	1.3	1.5	1.7	5.3	6.3	8.9	2.8	3.8	6.0
Naive	2.2	3.8	8.1	1.0	1.4	2.2	3.7	5.6	8.2	2.3	3.3	5.9
Instantaneous	2.2	3.8	7.8	1.0	1.4	1.9	4.0	5.9	8.9	2.4	3.4	5.9
Exp. W=10	4.9	5.8	7.3	1.3	1.4	1.5	6.8	7.7	10.3	4.2	4.8	6.3
Homogeneous	6.7	6.7	6.8	2.4	2.4	2.2	11.0	11.9	14.1	5.5	5.7	6.0